

NAUFAL AUZAN RAMADHAN

Engineer at Every Layer. Hardware to Code.

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PROFILE SUMMARY

Final-year Computer Engineering student at IPB University with a multidisciplinary background spanning embedded systems, robotics, mechanical fabrication, and full-stack software development. Proven track record of delivering end-to-end engineering solutions — from precision-welded UGV chassis to real-time teleoperation web interfaces and interactive 3D product platforms. Driven by the conviction that great engineering means following problems wherever they lead, regardless of domain.

EDUCATION

Diploma IV – Computer Engineering Technology | IPB University, Vocational School

August 2022 – Present (Expected Graduation: 2026) | GPA: 3.56 / 4.00

WORK EXPERIENCE

Engineering Research Intern | **Politeknik Enjiniring Pertanian Indonesia (PEPI)**

Sep 2025 – Dec 2025 | Serpong, Banten

- Conducted field research across laboratory and workshop facilities to identify technological gaps and development opportunities in agricultural engineering.
- Conceptualized a Modular Semi-Autonomous UGV (Raspberry Pi + RTK-GPS) capable of full autonomous operation and teleoperation — inspired by conventional tractor technology and redesigned for lighter weight, lower cost, and improved portability. Delivered as a 3D design prototype and system architecture blueprint.
- Designed a Semi-Autonomous Mini Combine Harvester integrating Raspberry Pi with a paddy-field detection algorithm for autonomous navigation assistance, supporting both automatic and teleoperation modes.

Research Assistant – Embedded Systems & Fabrication | **IPB University, Electromechanical Workshop**

Feb 2025 – Jun 2025 | Bogor, West Java

- Developed a remote-controlled UGV for precision anthracnose detection, integrating AI-based image classification with embedded hardware control systems.
- Executed full mechanical fabrication lifecycle — cutting, MIG welding, bending, drilling, and lathe operations — for a heavy-duty UGV chassis rated at 200 kg payload capacity.
- Contributed to embedded system integration and rapid prototyping across hardware and software layers.

Robotics Engineering Intern | **Roboratory Indonesia**

Feb 2021 – Jun 2021 | Bogor (Pre-university Internship)

- Designed and produced Arduino-based educational robotics kits integrating ultrasonic and infrared sensors with Scratch-based logic programming.
- Managed component inventory and quality-tested 30+ robotic kits, ensuring zero-defect delivery for educational deployment.

SELECTED PROJECTS

Robot Teleoperation Web Interface | **FastAPI · WebSockets · Leaflet.js · Vanilla JS**

- Built a real-time robot teleoperation dashboard functioning as a Hardware Abstraction Layer between browser UI and physical UGV hardware.
- Implemented live GPS tracking with Leaflet.js, bidirectional WebSocket command channels, and manual/autonomous mode switching.

PhytoPulse – Smart Greenhouse Dashboard | **React · Vite · Three.js · Zustand**

- Developed a full-stack IoT monitoring dashboard with live sensor data visualization and an animated 3D-rendered greenhouse model using Three.js.
- Architected global state management with Zustand, enabling real-time reactive UI updates across all sensor panels.

3D Product Configurator | [React](#) · [Three.js](#) · [Vite](#) · [TypeScript](#)

- Built an interactive 3D product visualization platform enabling real-time customization, exploded-view part inspection, and immersive product exploration.
- Designed for production-ready business use cases including e-commerce visualization and product education, leveraging Three.js for high-performance rendering.

Modular Semi-Autonomous UGV | [Raspberry Pi](#) · [RTK-GPS](#) · [Fusion360](#) · [Navigation Algorithm](#)

- A conceptual design for a lightweight, modular UGV built around Raspberry Pi and RTK-GPS, capable of switching between full autonomous operation and manual teleoperation.
- Inspired by conventional tractor technology but fundamentally redesigned — prioritizing portability, lower production cost, and higher field efficiency for agricultural environments. Delivered as a complete 3D prototype in Fusion360 alongside a system architecture blueprint.

Semi-Autonomous Mini Combine Harvester | [Raspberry Pi](#) · [RTK-GPS](#) · [Fusion360](#) · [Navigation Algorithm](#)

- A design concept for a semi-autonomous harvesting machine that integrates Raspberry Pi with a paddy field detection algorithm, enabling automated navigation assistance across unstructured agricultural terrain.
- The system supports seamless switching between autonomous and teleoperation modes, with the full mechanical structure and component layout visualized as a 3D model in Fusion360.

Remote-Controlled UGV for Anthracnose Detection | [NVIDIA Jetson](#) · [ESP32](#) · [YOLOv8](#) · [SMAW Welding](#)

- This project integrates an NVIDIA Jetson-based computer vision module with a hand-welded steel chassis, bridging mechanical fabrication and embedded AI into a single field-deployable system.
- Trained on a curated chili pepper dataset, the YOLOv8 model enables real-time anthracnose disease detection directly in field conditions — combining SMAW welding, ESP32 motor control, and OpenCV inference into one cohesive pipeline.

Heavy-Duty UGV Chassis Development | [SMAW Welding](#) · [MIG Welding](#) · [Metal Fabrication](#)

- Engineered and fabricated a heavy-duty steel chassis for unmanned ground vehicle applications, rated for payloads up to 200 kg. The design prioritized structural integrity, balanced weight distribution, and terrain adaptability.

Adaptive UGV Camera Mast System | [Camera](#) · [Servo Motor](#) · [Mechanical Design](#)

- A custom-designed camera mast prototype built to enhance environmental perception for Unmanned Ground Vehicles. The system integrates a three-camera configuration — front, left, and right — delivering comprehensive 270° visual coverage to support real-time navigation and object detection in dynamic field conditions.
- Designed with adaptability in mind, the mast structure allows for modular camera repositioning depending on mission requirements.

IoT-Based Attendance System | [ESP32](#) · [NFC](#) · [MySQL](#)

- Built an NFC-based attendance system using ESP32 with cloud data sync to a MySQL backend; handled full hardware assembly and quality assurance.

Automatic Solar Panel Tracking System | [Embedded Systems](#) · [Mechanical Fabrication](#)

- Served as Mechanical PIC, overseeing full fabrication — welding, cutting, drilling, and lathe operations — for a dual-axis proportional tracking mechanism.

Automatic Portable Water Dispenser | [GAL IC](#) · [Infrared Sensor](#) · [Fusion360](#) · [3D Printing](#)

- Designed sensor-triggered dispensing logic, modeled the enclosure in Fusion360, and produced components via 3D printing and custom PCB fabrication.

PiggyNote – Personal Finance Tracker | [HTML](#) · [CSS](#) · [JavaScript](#) · [Bootstrap](#) · [Java](#) · [Android](#)

- Delivered a dual-platform income/expense tracker: web version with Vanilla HTML/CSS/JS and native Android version built in Java with Android Studio.

SKILLS

Programming & Web: Python, JavaScript, TypeScript, React.js, PHP, Laravel, Java, HTML, CSS

Embedded & IoT: Arduino, ESP32, Raspberry Pi, MQTT, WebSockets, RTK-GPS

Design & Simulation: Fusion360, Fritzing, Proteus, Eagle, Three.js

Fabrication: SMAW / MIG Welding, Lathe, Metal Fabrication, 3D Printing

Machine Learning: NumPy, Pandas, Matplotlib, Jupyter Notebook

Database & Networking: MySQL, PostgreSQL, Cisco Equipment, VLAN, DHCP, DNS

Languages: English – IELTS 7.0 (Professional Proficiency) | Indonesian – Native